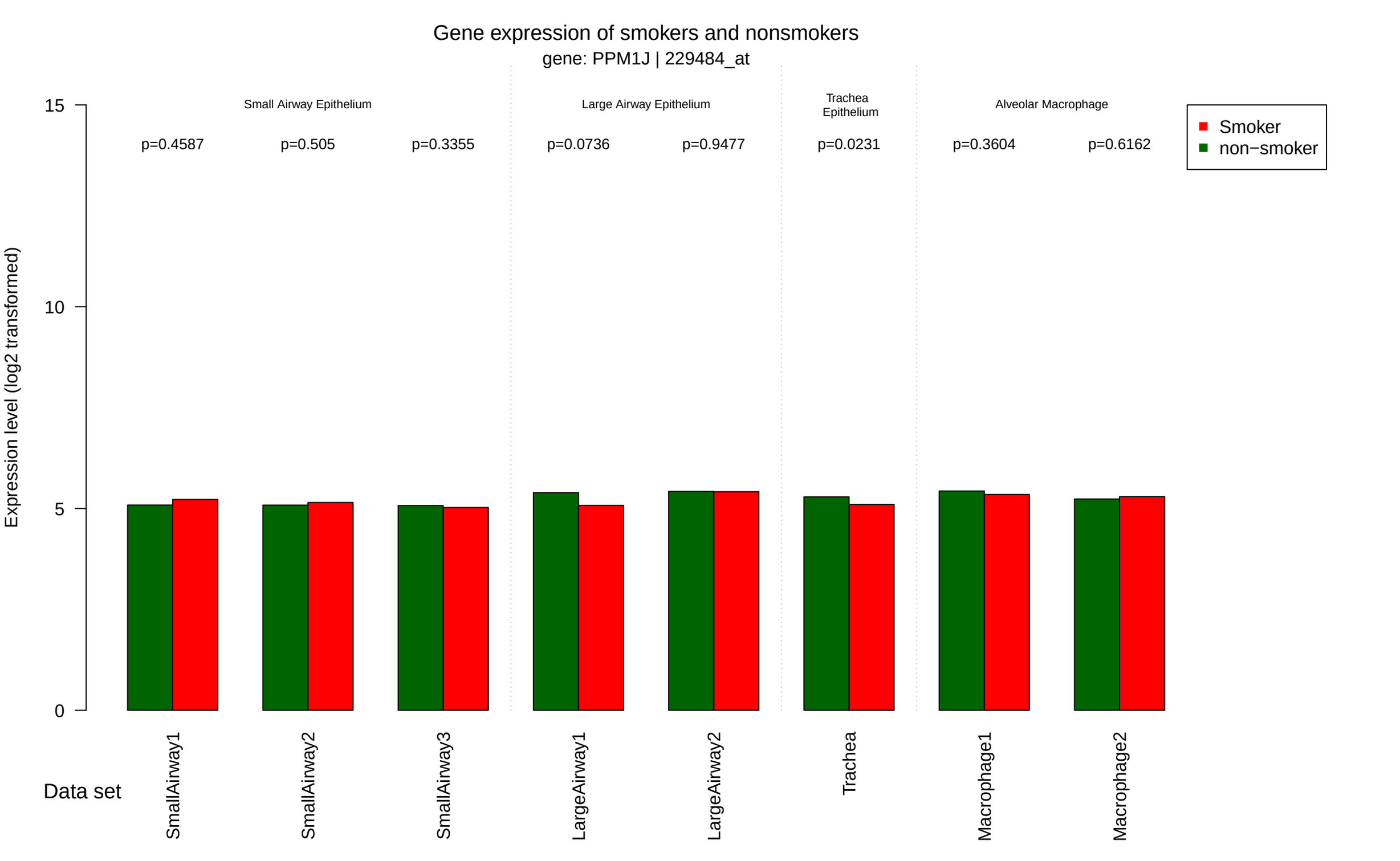
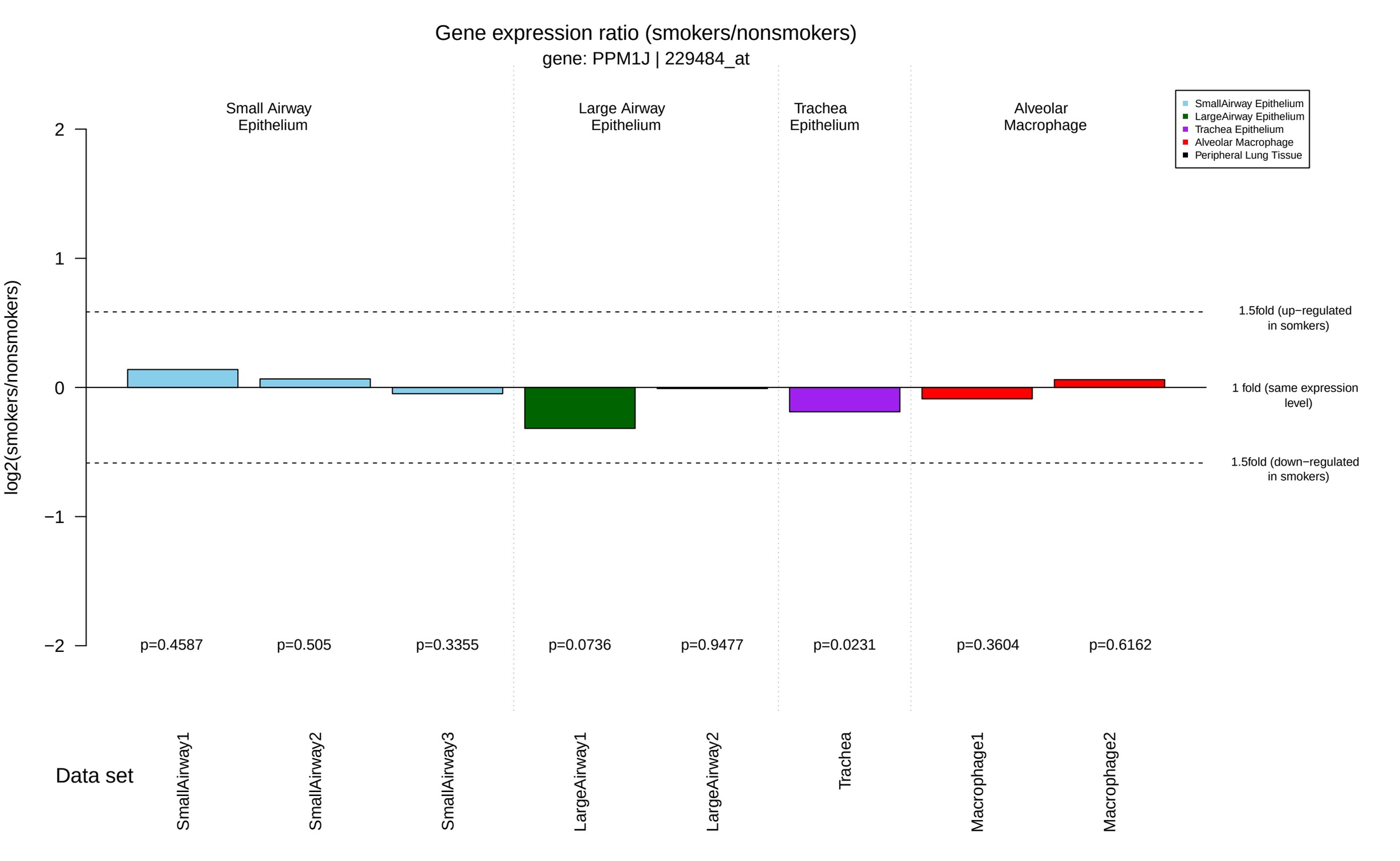
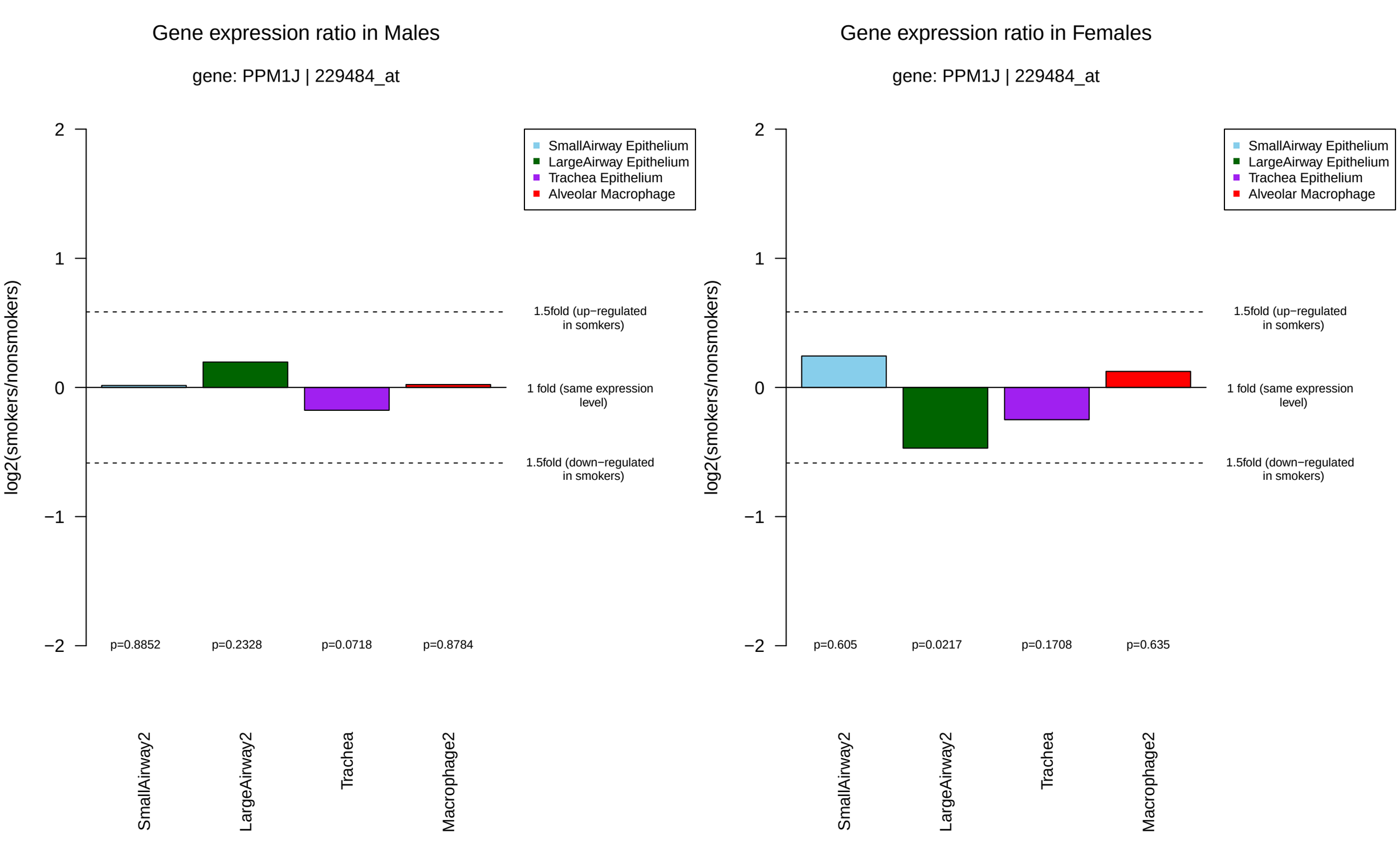
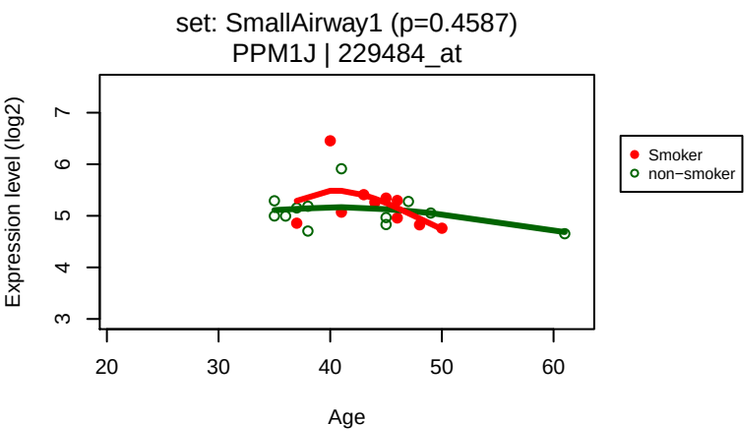
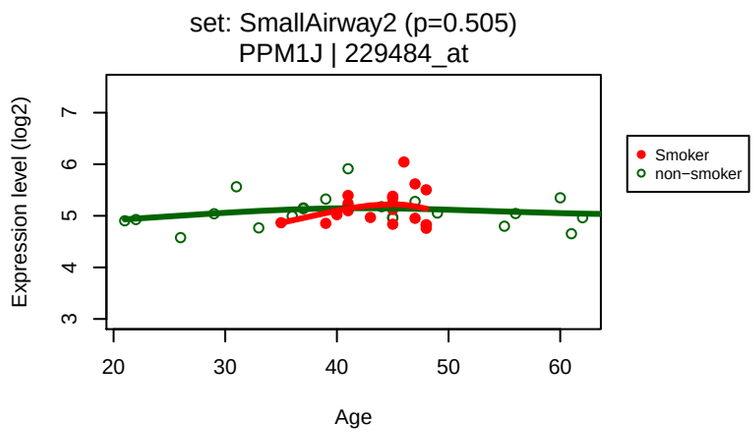
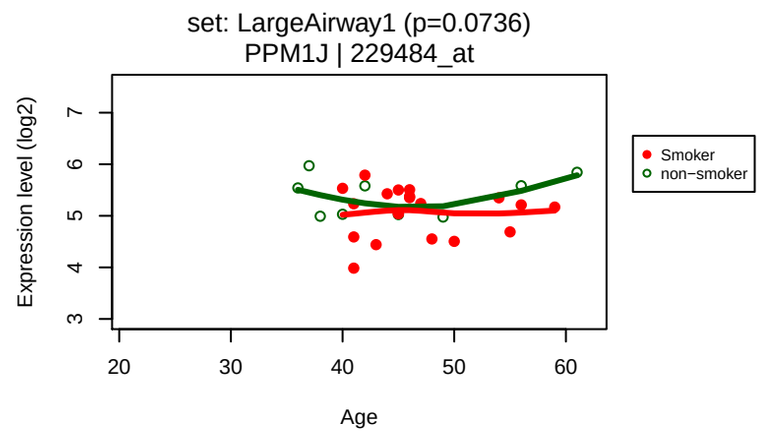

Gene expression of smokers and nonsmokers
gene: PPM1J | 229484_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4587
p=0.505
p=0.3355
p=0.0736
p=0.9477
p=0.0231
p=0.3604
p=0.6162
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

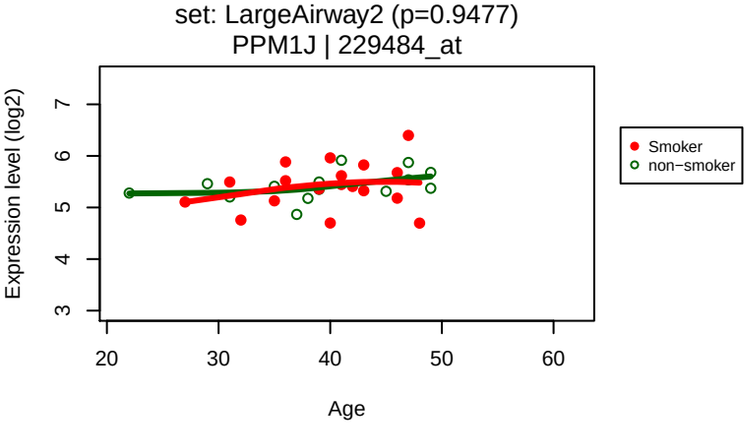

Gene expression ratio (smokers/nonsmokers)
gene: PPM1J | 229484_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4587
p=0.505
p=0.3355
p=0.0736
p=0.9477
p=0.0231
p=0.3604
p=0.6162
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

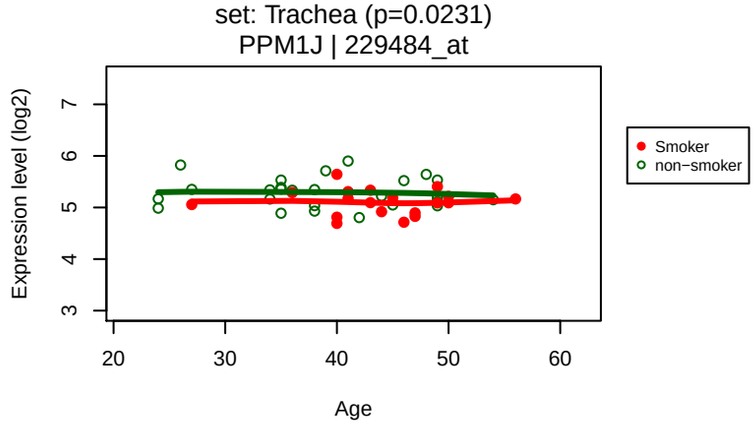

Gene expression ratio in Males
gene: PPM1J | 229484_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8852
p=0.2328
p=0.0718
p=0.8784
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1J | 229484_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.605
p=0.0217
p=0.1708
p=0.635
SmallAirway2
LargeAirway2
Trachea
Macrophage2

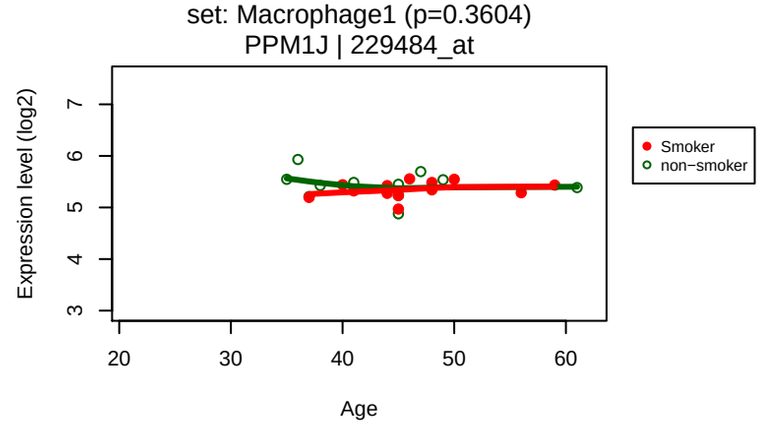

set: SmallAirway1 (p=0.4587)
PPM1J | 229484_at
Expression level (log2)
Age
Smoker
non−smoker

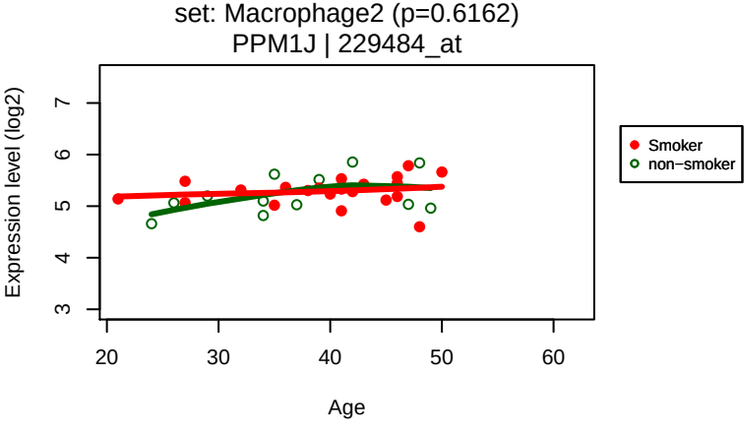

set: SmallAirway2 (p=0.505)
PPM1J | 229484_at
Expression level (log2)
Age
Smoker
non−smoker

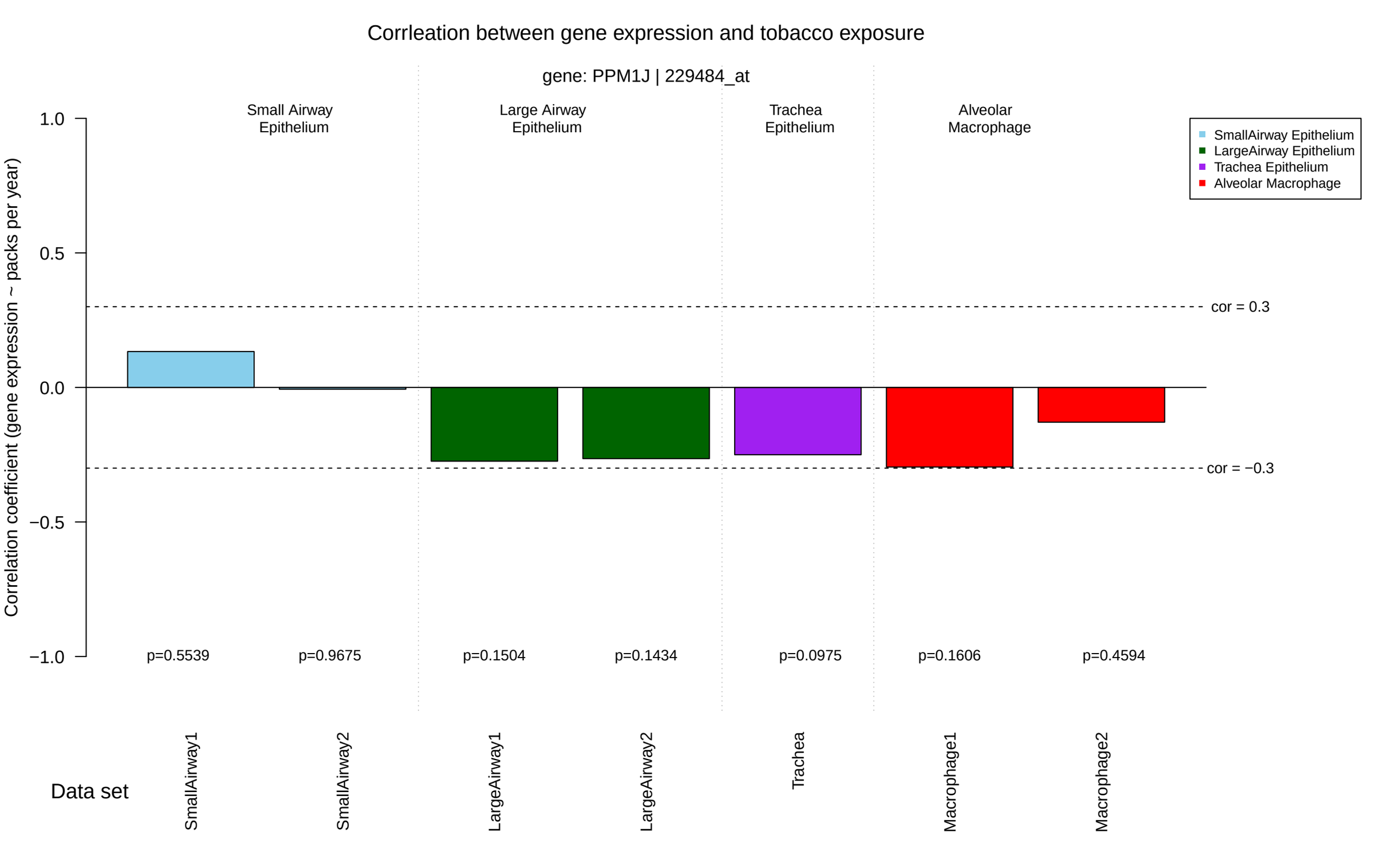

set: LargeAirway1 (p=0.0736)
PPM1J | 229484_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9477)
PPM1J | 229484_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0231)
PPM1J | 229484_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3604)
PPM1J | 229484_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6162)
PPM1J | 229484_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1J | 229484_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5539
p=0.9675
p=0.1504
p=0.1434
p=0.0975
p=0.1606
p=0.4594
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2